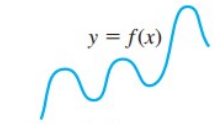
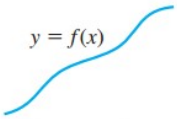
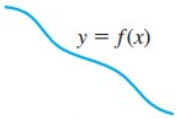
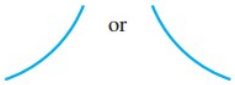
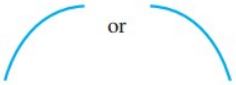
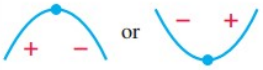
EXAMPLE 1 Use the guidelines to sketch the curve $y = \dfrac{2x^2}{x^2 - 1}$.

EXAMPLE 9 Sketch the graph of $f(x) = \dfrac{(x + 1)^2}{1 + x^2}$.

EXAMPLE 10 Sketch the graph of $f(x) = \dfrac{x^2 + 4}{2x}$.

Graphical Behavior of Functions from Derivatives

 Differentiable ⇒ smooth, connected; graph may rise and fall	 $y' > 0 \Rightarrow$ rises from left to right; may be wavy	 $y' < 0 \Rightarrow$ falls from left to right; may be wavy
 $y'' > 0 \Rightarrow$ concave up throughout; no waves; graph may rise or fall or both	 $y'' < 0 \Rightarrow$ concave down throughout; no waves; graph may rise or fall or both	 y'' changes sign at an inflection point
 y' changes sign ⇒ graph has local maximum or local minimum	 $y' = 0$ and $y'' < 0$ at a point; graph has local maximum	 $y' = 0$ and $y'' > 0$ at a point; graph has local minimum

Sketching the General Shape, Knowing y'

Each of Exercises 51–72 gives the first derivative of a continuous function $y = f(x)$. Find y'' and then use Steps 2–4 of the graphing procedure on page 207 to sketch the general shape of the graph of f.

51. $y' = 2 + x - x^2$

52. $y' = x^2 - x - 6$

53. $y' = x(x - 3)^2$

54. $y' = x^2(2 - x)$

55. $y' = x(x^2 - 12)$

56. $y' = (x - 1)^2(2x + 3)$

57. $y' = (8x - 5x^2)(4 - x)^2$

58. $y' = (x^2 - 2x)(x - 5)^2$

59. $y' = \sec^2 x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$

60. $y' = \tan x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$

61. $y' = \cot \frac{\theta}{2}, \quad 0 < \theta < 2\pi$

62. $y' = \csc^2 \frac{\theta}{2}, \quad 0 < \theta < 2\pi$

63. $y' = \tan^2 \theta - 1, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$

64. $y' = 1 - \cot^2 \theta, \quad 0 < \theta < \pi$

65. $y' = \cos t, \quad 0 \le t \le 2\pi$

66. $y' = \sin t, \quad 0 \le t \le 2\pi$

67. $y' = (x + 1)^{-2/3}$

68. $y' = (x - 2)^{-1/3}$

Graphing Rational Functions

Graph the rational functions in Exercises 77–94 using all the steps in the graphing procedure on page 207.

77. $y = \dfrac{2x^2 + x - 1}{x^2 - 1}$

78. $y = \dfrac{x^2 - 49}{x^2 + 5x - 14}$

79. $y = \dfrac{x^4 + 1}{x^2}$

80. $y = \dfrac{x^2 - 4}{2x}$

81. $y = \dfrac{1}{x^2 - 1}$

82. $y = \dfrac{x^2}{x^2 - 1}$

83. $y = -\dfrac{x^2 - 2}{x^2 - 1}$

84. $y = \dfrac{x^2 - 4}{x^2 - 2}$

85. $y = \dfrac{x^2}{x + 1}$

86. $y = -\dfrac{x^2 - 4}{x + 1}$

87. $y = \dfrac{x^2 - x + 1}{x - 1}$

88. $y = -\dfrac{x^2 - x + 1}{x - 1}$

89. $y = \dfrac{x^3 - 3x^2 + 3x - 1}{x^2 + x - 2}$

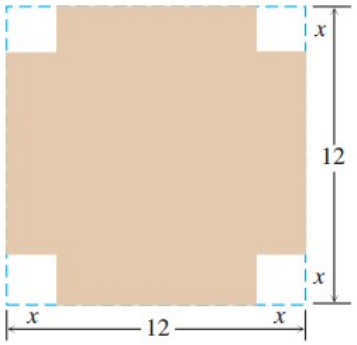
90. $y = \dfrac{x^3 + x - 2}{x - x^2}$

4.5 Applied Optimization

Solving Applied Optimization Problems

1. *Read the problem*. Read the problem until you understand it. What is given? What is the unknown quantity to be optimized?
2. *Draw a picture*. Label any part that may be important to the problem.
3. *Introduce variables*. List every relation in the picture and in the problem as an equation or algebraic expression, and identify the unknown variable.
4. *Write an equation for the unknown quantity*. If you can, express the unknown as a function of a single variable or in two equations in two unknowns. This may require considerable manipulation.
5. *Test the critical points and endpoints in the domain of the unknown*. Use what you know about the shape of the function's graph. Use the first and second derivatives to identify and classify the function's critical points.

EXAMPLE 1 An open-top box is to be made by cutting small congruent squares from the corners of a 12-in.-by-12-in. sheet of tin and bending up the sides. How large should the squares cut from the corners be to make the box hold as much as possible?



Examples from Mathematics and Physics

EXAMPLE 3 A rectangle is to be inscribed in a semicircle of radius 2. What is the largest area the rectangle can have, and what are its dimensions?

EXAMPLE 1 A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

V EXAMPLE 2 A cylindrical can is to be made to hold 1 L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.

V EXAMPLE 3 Find the point on the parabola $y^2 = 2x$ that is closest to the point $(1, 4)$.

INDETERMINATE FORMS AND L'HOSPITAL'S RULE

L'HOSPITAL'S RULE Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \to a} f(x) = 0 \qquad \text{and} \qquad \lim_{x \to a} g(x) = 0$$

or that

$$\lim_{x \to a} f(x) = \pm\infty \qquad \text{and} \qquad \lim_{x \to a} g(x) = \pm\infty$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or ∞/∞.) Then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is ∞ or $-\infty$).

EXAMPLE 1 Find $\lim_{x \to 1} \frac{\ln x}{x - 1}$.

EXAMPLE 2 Calculate $\lim_{x \to \infty} \frac{e^x}{x^2}$.

EXAMPLE 3 Calculate $\lim_{x \to \infty} \frac{\ln x}{\sqrt[3]{x}}$.

EXAMPLE 5 Find $\lim_{x \to \pi^-} \frac{\sin x}{1 - \cos x}$.

INDETERMINATE PRODUCTS

If $\lim_{x \to a} f(x) = 0$ and $\lim_{x \to a} g(x) = \infty$ (or $-\infty$), then it isn't clear what the value of $\lim_{x \to a} f(x)g(x)$, if any, will be. There is a struggle between f and g. If f wins, the answer will be 0; if g wins, the answer will be ∞ (or $-\infty$). Or there may be a compromise where the answer is a finite nonzero number. This kind of limit is called an **indeterminate form of type $0 \cdot \infty$**. We can deal with it by writing the product fg as a quotient:

$$fg = \frac{f}{1/g} \quad \text{or} \quad fg = \frac{g}{1/f}$$

This converts the given limit into an indeterminate form of type $\frac{0}{0}$ or ∞/∞ so that we can use l'Hospital's Rule.

V EXAMPLE 6 Evaluate $\lim_{x \to 0^+} x \ln x$.

INDETERMINATE DIFFERENCES

If $\lim_{x \to a} f(x) = \infty$ and $\lim_{x \to a} g(x) = \infty$, then the limit

$$\lim_{x \to a} [f(x) - g(x)]$$

EXAMPLE 7 Compute $\lim_{x \to (\pi/2)^-} (\sec x - \tan x)$.

INDETERMINATE POWERS

Several indeterminate forms arise from the limit

$$\lim_{x \to a} [f(x)]^{g(x)}$$

1. $\lim_{x \to a} f(x) = 0$ and $\lim_{x \to a} g(x) = 0$ type 0^0
2. $\lim_{x \to a} f(x) = \infty$ and $\lim_{x \to a} g(x) = 0$ type ∞^0
3. $\lim_{x \to a} f(x) = 1$ and $\lim_{x \to a} g(x) = \pm\infty$ type 1^∞

Each of these three cases can be treated either by taking the natural logarithm:

$$\text{let} \quad y = [f(x)]^{g(x)}, \quad \text{then} \quad \ln y = g(x) \ln f(x)$$

or by writing the function as an exponential:

$$[f(x)]^{g(x)} = e^{g(x) \ln f(x)}$$

EXAMPLE 8 Calculate $\lim_{x \to 0^+} (1 + \sin 4x)^{\cot x}$.

EXAMPLE 9 Find $\lim_{x \to 0^+} x^x$.

Review Exercises

$$\lim_{x\to-\infty}\frac{\sqrt{9x^2-2}}{6x+1}=$$

$$\lim_{x\to\infty}\frac{2\sin(x^5)}{x^4+1}$$

$$\lim_{x\to2}\frac{x^2-4}{3x|x-2|}$$

$$\lim_{x\to\infty}\left[\ln(x^2+1)-\ln(x^2-1)\right]$$

$$\lim_{x\to0}\left[\frac{\sqrt{1+2x}-\sqrt{1-4x}}{x}\right]=$$

$$\lim_{x\to-\infty}\frac{\sqrt{9x^2-2}}{6x+1}=$$

$$\lim_{x\to1}\frac{\sqrt{x+3}-2}{x-1}=$$

$$\lim_{x\to2}\frac{\sqrt{x+2}-\sqrt{2x}}{x^2-2x}$$

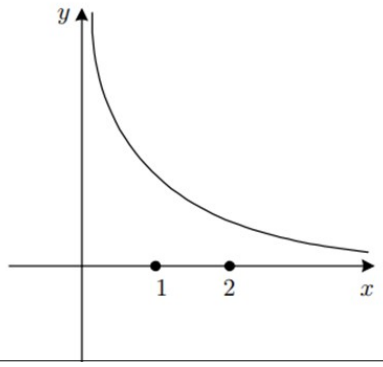
$$\lim_{x\to\infty}x\tan\frac{1}{x}$$

$$\lim_{x\to-\infty}\frac{\sqrt{x^2+4}}{3x-2}$$

(8 - points) Given that $\lim_{x\to 2}\left(3x-\frac{2}{5}\right)=\frac{28}{5}$ and $\epsilon = 0.009$. Find the largest possible value of δ that satisfies the conditions given in the $\epsilon-\delta$ definition of a limit.

5. **[8 points]** Using the ϵ, δ definition of limit, prove that $\lim_{x\to 1}\left(-1+\frac{3}{2}x\right)=\frac{1}{2}$

6. (10-points) Use the graph of $f(x)=\frac{1}{x}$ to find the largest number δ such that if $|x-1|<\delta$, then $|f(x)-1|<0.1$. (Show your work and write your answer in simplest rational form $\frac{p}{q}$).



Q.3 (10 pts) Find the number $a \neq 0$ such that the function

$$f(x) = \begin{cases} \dfrac{\sin(ax)}{x} + \cos x & \text{for} \quad x < 0 \\ x^2 + 3e^{2x} & \text{for} \quad x \geq 0 \end{cases}$$

is continuous **everywhere.**

If $f(x) = \begin{cases} \dfrac{\sin 2x}{\sin 3x} & \text{if} \quad x < 0 \\ x^2 + a & \text{if} \quad 0 \leq x < 1 \\ x - 2b & \text{if} \quad x \geq 1 \end{cases}$

is continuous everywhere, then $a + b =$

(c)[10] Let $f(x) = \begin{cases} \dfrac{\sqrt{ax+b} - 2}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

Determine the values of the real numbers a and b so that $f(x)$ is continuous at the point $x = 0$.

If $x^2 - 3 \leq 3f(x) + 10 \leq -2x - 4$, then $\lim_{x \to -1} f(x) =$

(a) By definition, the limit $\lim_{h \to 0} \dfrac{\sqrt[4]{16+h} - 2}{h}$ represents the derivative of a function $f(x)$ at the number $x = x_0$. Find $f(x)$ and x_0.

Using the Intermediate Value Theorem, we conclude that the equation

$$x^5 = 2x^3 + 2$$

has at least one root in the interval

(a) $(0, 2)$ (correct)
(b) $(0, 1)$
(c) $(-1, 1)$
(d) $(-1, 0)$
(e) $(2, 3)$

Q.1 (15 pts) Show that the equation

$$\arctan x = 2 - x - x^3$$

has exactly ONE solution.

Q.3 (10 pts) Let $f(x) = \begin{cases} x^2 + \arcsin x & \text{for } x \geq 0 \\ x + e^{x^2} & \text{for } x < 0 \end{cases}$. Find $f'(0)$ if it exists, or explain why it does NOT exist.

If a and b are the values that make $f(x) = \begin{cases} ax^3 & x \leq 2 \\ x^2 + b & x > 2 \end{cases}$ differentiable, then $a + b =$

Exercise 1. Use linear approximation to approximate $(1.001)^{1000}$

A linearization of the function $f(x) = e^x + \sin x$ at $x = 0$ is

Using a linear approximation (or differentials), the best estimation to $\sqrt[3]{1001}$ is $10 + B$. Then $B =$

By using the linear approximation of $f(x) = \sqrt{100 + x}$ at $a = 0$, $\sqrt{100.5}$ is approximately equal to

If $L(x)$ is the linearization of $f(x) = 2 + \ln(1 - 4x)$ at $a = 0$, then $L(-1) =$

(10 pnts) a) Write the definition of the derivative $f'(a)$ of a function $f(x)$ at $x = a$.
b) Use the definition to prove that the derivative of the function $f(x) = 3x^2 - 2$ at $x = 2$ equals 12.
c) Find the equation of the tangent line to the graph of $f(x)$ at the point $(2, 10)$.

(a) By definition, the limit $\lim_{h \to 0} \dfrac{\sqrt[4]{16+h} - 2}{h}$ represents the derivative of a function $f(x)$ at the number $x = x_0$. Find $f(x)$ and x_0.

Q.2 (10 pts) For what values of the numbers a and b is the curve $f(x) = e^{ax^2+bx}$ **tangent** to the line $y = e$ at the point $(x, y) = (1, e)$?

(8 - points) Use the Intermediate Value Theorem to show that the graphs of the functions $f(x) = \sqrt{x}$ and $g(x) = \cos x$ intersect on the interval $\left[0, \frac{\pi}{2}\right]$.

Let $f(x) = x^3 - 3x + 2$ for $-2 \leq x \leq 2$. Then, the numbers that satisfy the conclusion of the Mean Value Theorem are

Consider the function $f(x) = x + e^x$. The value of c that satisfies the conclusion of the **Mean Value Theorem** on the interval $[0, 1]$ is

If $f(4) = -1$ and $f'(x) \geq 5$ for $2 \leq x \leq 4$, then the largest possible value of $f(2)$ is

for f

Let $a > 0$ and let $f(x) = x^2 + \frac{x}{a}$, $a \leq x \leq 2a$. The value of a, that makes the average rate of change of the function f on $[a, 2a]$ the smallest possible, is

The radius of a circle increases from 3 cm to 3.025 cm. Using differentials, the best approximation in the increase of its area is equal to

(a) $0.15\,\pi$ cm^2

(b) $0.75\,\pi$ cm^2

(c) $0.45\,\pi$ cm^2

(d) $0.09\,\pi$ cm^2

(e) $0.18\,\pi$ cm^2

Q.6 (10 pts) A particle moves along the vertical line $x = 4$ with its $y-$coordinate increasing at a rate of 2 cm/sec. How fast is the **distance**, say S, of the particle to the **origin** changing when the particle is at the point $(4, 3)$?

The volume of a metal cube is increasing at a rate of $30\,cm^3/h$ when the side is $10\,cm$. The rate at which the side is increasing at that moment is

The volume of a metal cube is increasing at a rate of $30\,cm^3/h$ when the side is $10\,cm$. The rate at which the side is increasing at that moment is

If a circular plate of metal is heated in an oven, its radius increases at the rate of 0.01 cm/min. When the radius is 50 cm, the plate's area is increasing with rate

(b) Find $f'(x) = \dfrac{dy}{dx}$ if $y = f(x) = \dfrac{x^{\pi} + \cos x}{1 + x^3}$.

(c) Find $f'(x)$ if $f(x) = (\cos x + 3)^{\sqrt{2 + \sin x}}$.

(b) Calculate y' at the point $(x, y) = (1, -1)$ if $y = y(x)$ and $x^4 + y^4 + 2xy = 0$.

Consider the equation $x^y = y^x$ the rate of change of y with respect to x at the point $(1, 1)$ is

The slope of the tangent line to the curve $x^2 - xy - y^2 = 1$ at the point $(2, 1)$ is

The slope of the line tangent to the curve $2x + \tan^{-1}(xy) = x^2 y^3 + 2$ at the point $(1, 0)$ is

An equation of the tangent line to the curve of

$$y^2 - x = \sin(4x^2 - 4)$$

at the point $(1, 1)$ is

Consider the implicit relation defined by $x \sin(xy - y^2) = x^2 - 1$.
[10] Find $\frac{dy}{dx}$.

If $f(t) = te^{2t} + 3^{2t+1}$, then $f'\left(\frac{1}{2}\right) =$

$$\frac{d}{dx}\left(x \sin^{-1} x + \sqrt{1-x^2}\right) =$$

If $f(x) = e^{\sin^2(\pi x)}$, then $f'(x) =$

If $f(x) = \dfrac{(x+1)^3(x-2)\cos x}{3^x(2x+1)}$, then $f'(0) =$

The sum of all critical numbers of the function $f(x) = \dfrac{x^2 + 14}{\sqrt{4x+1}}$ is

The graph of $f(x) = \dfrac{1}{2}x - \sin x, \quad 0 < x < 3\pi$ is concave upward on the interval(s)

(a) $(0, \pi)$ and on $(2\pi, 3\pi)$

(b) $\left(\dfrac{\pi}{2}, \dfrac{3\pi}{2}\right)$

(c) $\left(0, \dfrac{\pi}{2}\right)$ and on $\left(\pi, \dfrac{3\pi}{2}\right)$

(d) $\left(0, \dfrac{\pi}{2}\right)$ and on $(\pi, 3\pi)$

(e) $\left(\dfrac{3\pi}{2}, 3\pi\right)$

The graph of the function

$$f(x) = \frac{5}{2}x^{\frac{2}{3}} - x^{\frac{5}{3}}$$

is concave upward on the interval

(a) $(-\infty, -\frac{1}{2})$

(b) $(-\infty, 0)$

(c) $(-3, 4)$

(d) $(-2, \infty)$

(e) $(-7, 3)$

Let $f(x) = \frac{1}{\ln x}$. (a) Find the domain of f.

(b) Find the intervals of increase and decrease.

(c) Find the inflection point and the intervals where f is concave up and down.

(d) Determine the vertical and horizontal asymptotes.

(e) Sketch the graph of f.

EXAMPLE 3 Sketch the graph of $f(x) = xe^x$.

